AIVANCIN

Your next role starts with the work you already do.

For Malaysian graduates in jobs below their training: describe what you actually do, confirm what AI suggests, and see the closest graduate roles and a free first plan.

Team DOVUN · German-Malaysian Institute
SDC Hackathon 2026 · Domain One: The Aiversecity Future-Ready Workforce

The problem

40.7%

of employed Malaysian graduates aged 25 to 34 work in semi-skilled or low-skilled jobs.

751,100 people

Before spending evenings and savings on retraining, they need to know: **which graduate job is closest to the work I already do?**

Source: DOSM, Quarterly Skills-Related Underemployment by Age, Q2 2026 (CC BY 4.0)

Meet Hafiz (a synthetic persona)

27 · Computing diploma · Delivery van driver

His job title says

Vehicle-operator roles

- Light Truck Drivers
- Supervisors of vehicle operators, crane operators

0 of the top 10 roles are in computing

His weekends say

- Fixes friends' laptops and home wifi
- Installs software, cleans out viruses
- Backs up phones and laptops
- Keeps his uncle's shop website updated

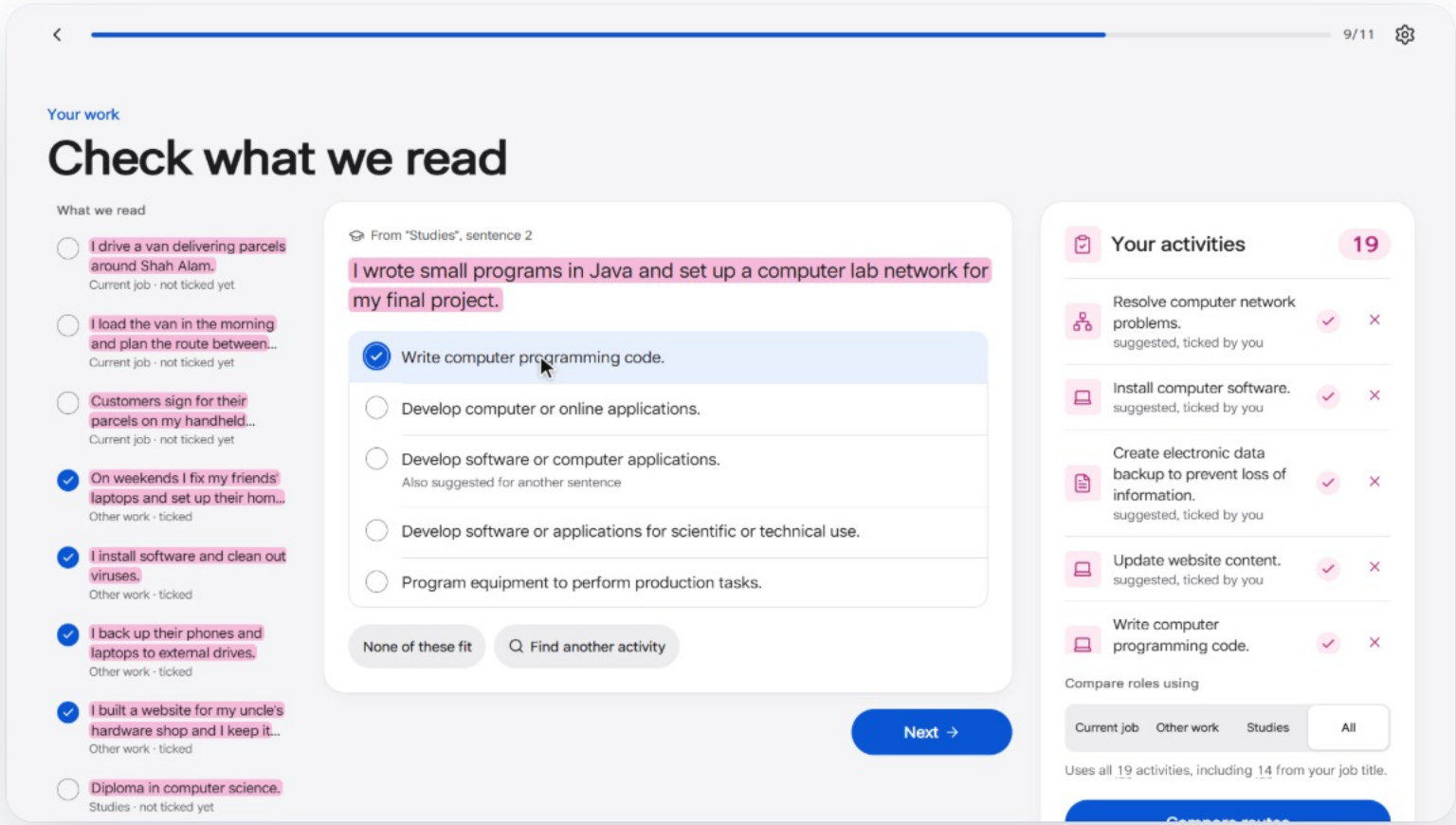
With that work confirmed: at least 6 of the top 10

His title hides this experience. Confirm it, then compare destinations.

Live prototype · Describe and confirm

Confirm what is true

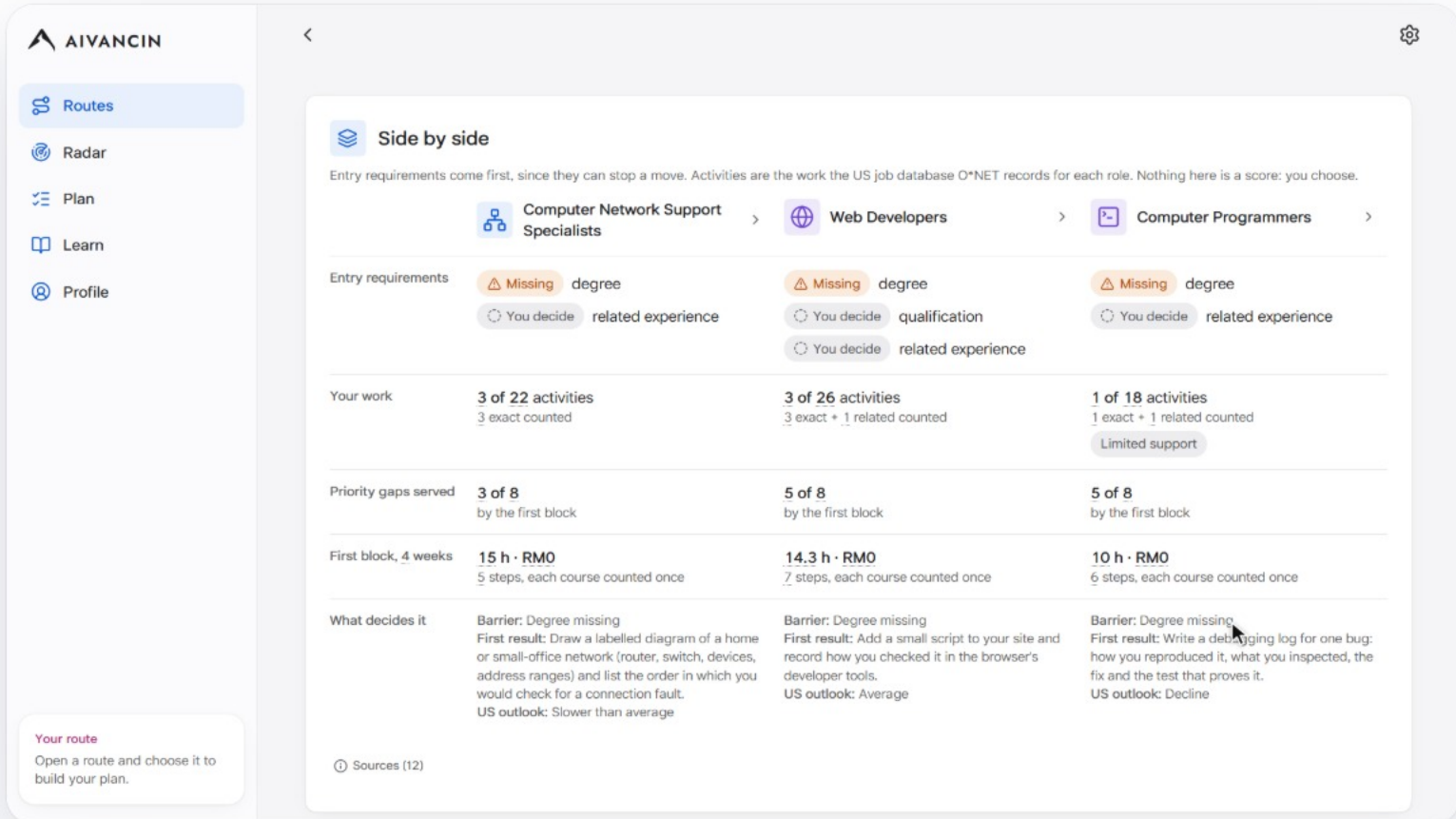
For each sentence, the AI model suggests matching work activities from O*NET's 2,087 and highlights the words behind them. Nothing counts until you tick it.



Live prototype · Compare

Routes that use your work

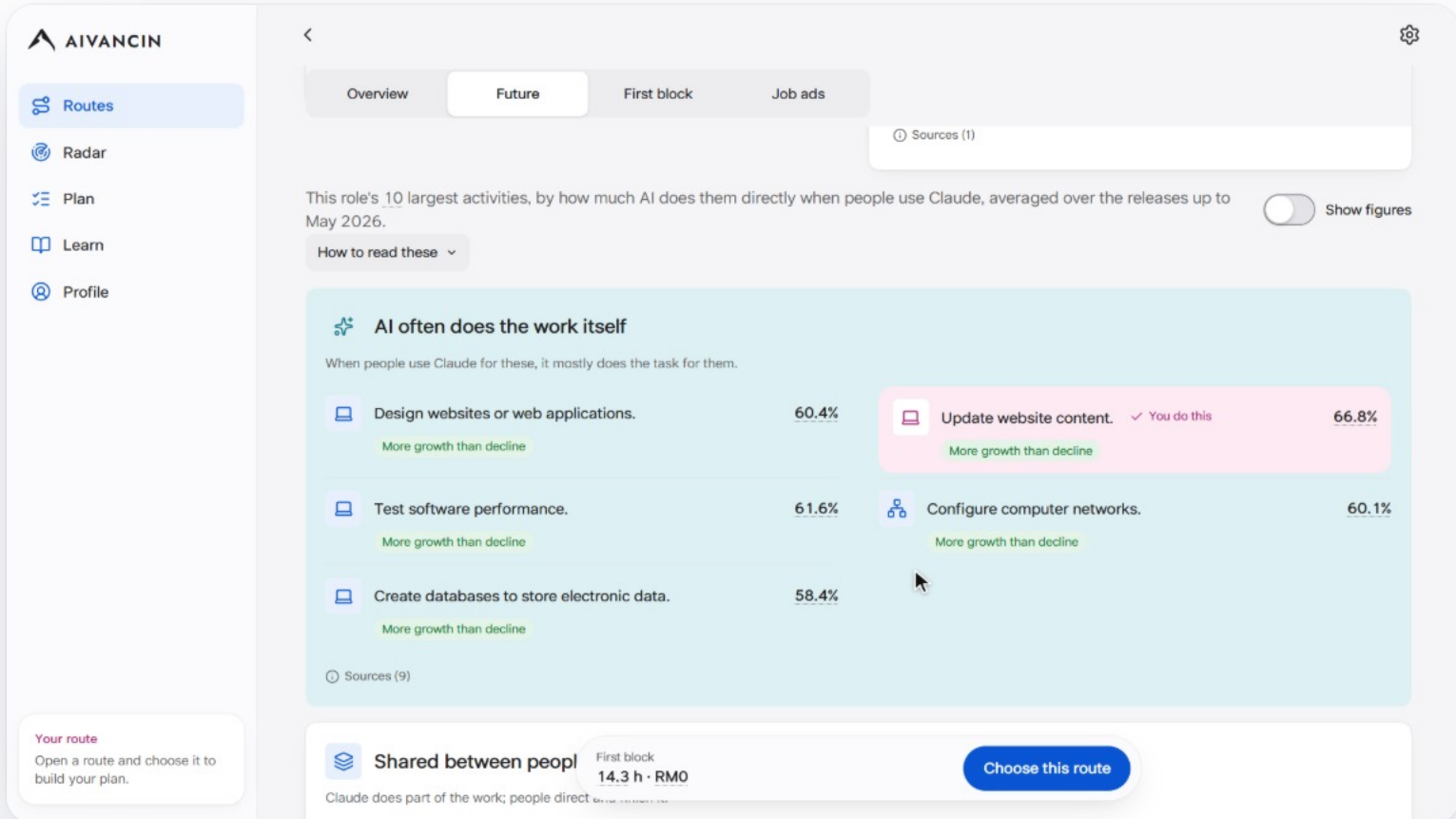
Roles ranked by shared work, side by side: what you already do, entry barriers still open, and the gaps a free first block covers. Reorder by outlook or decent work.



Live prototype · Route detail

The future of each route

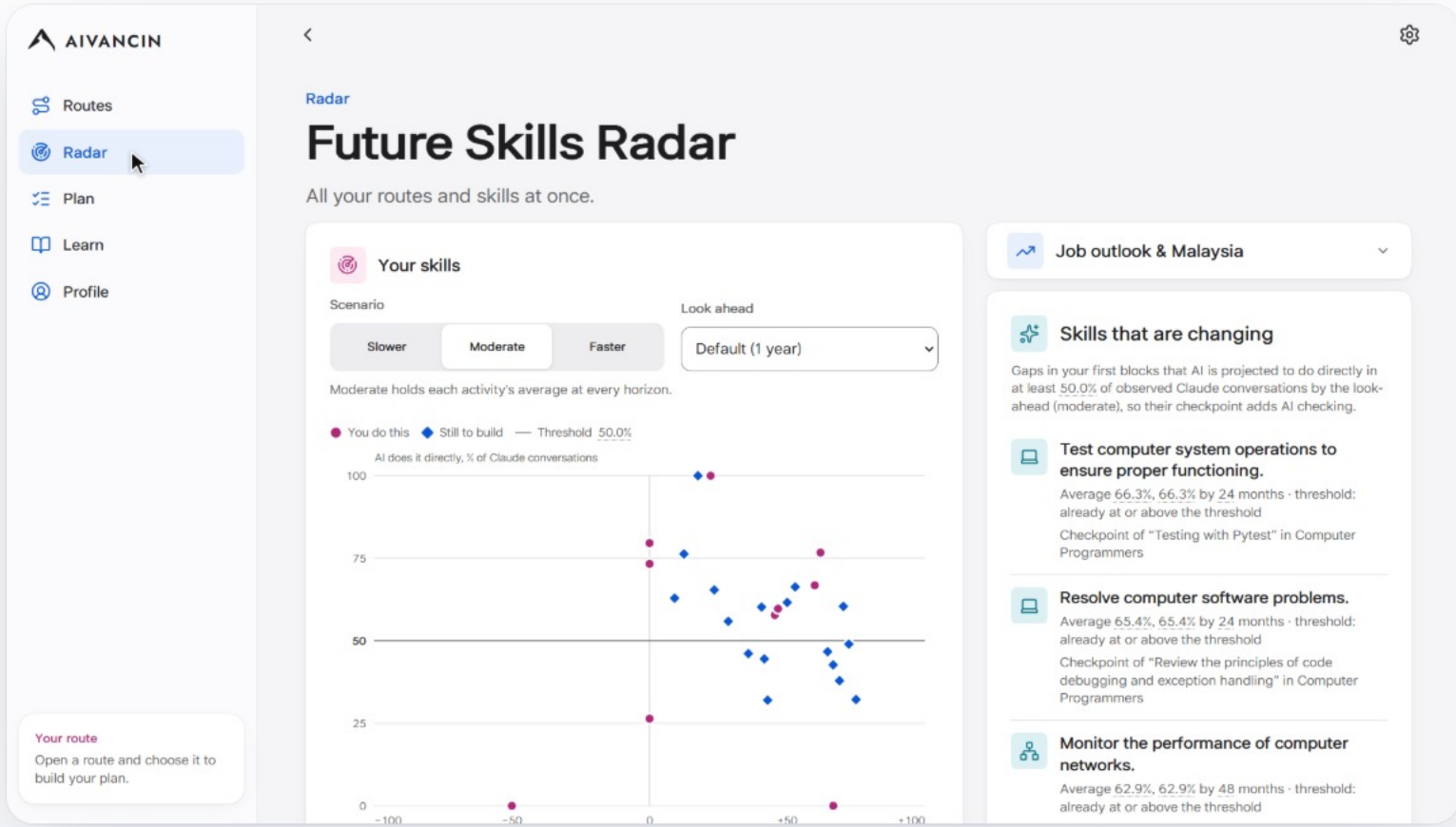
Three dated sources, kept separate: US job projections (BLS), Malaysia's observed trend for the job group (DOSM), and how often Claude already does each activity (Anthropic Economic Index).



Live prototype · Radar

Future Skills Radar

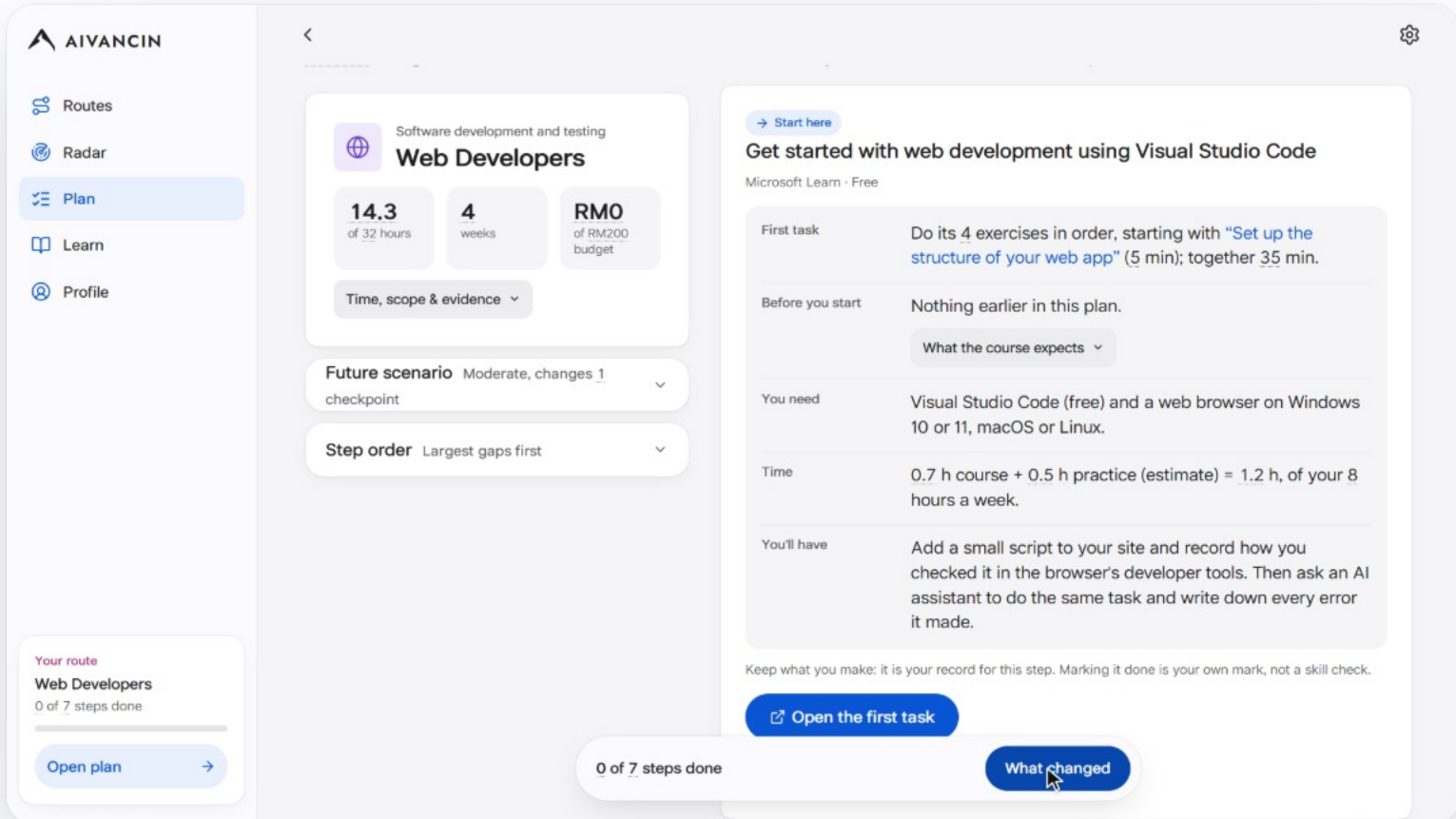
Your skills in pink, your gaps in blue. Right: more tied to growing US jobs. Up: Claude already does more of that work. Switch slower, moderate or faster AI adoption; every projection is an estimate.



Live prototype · Plan

A free first learning block

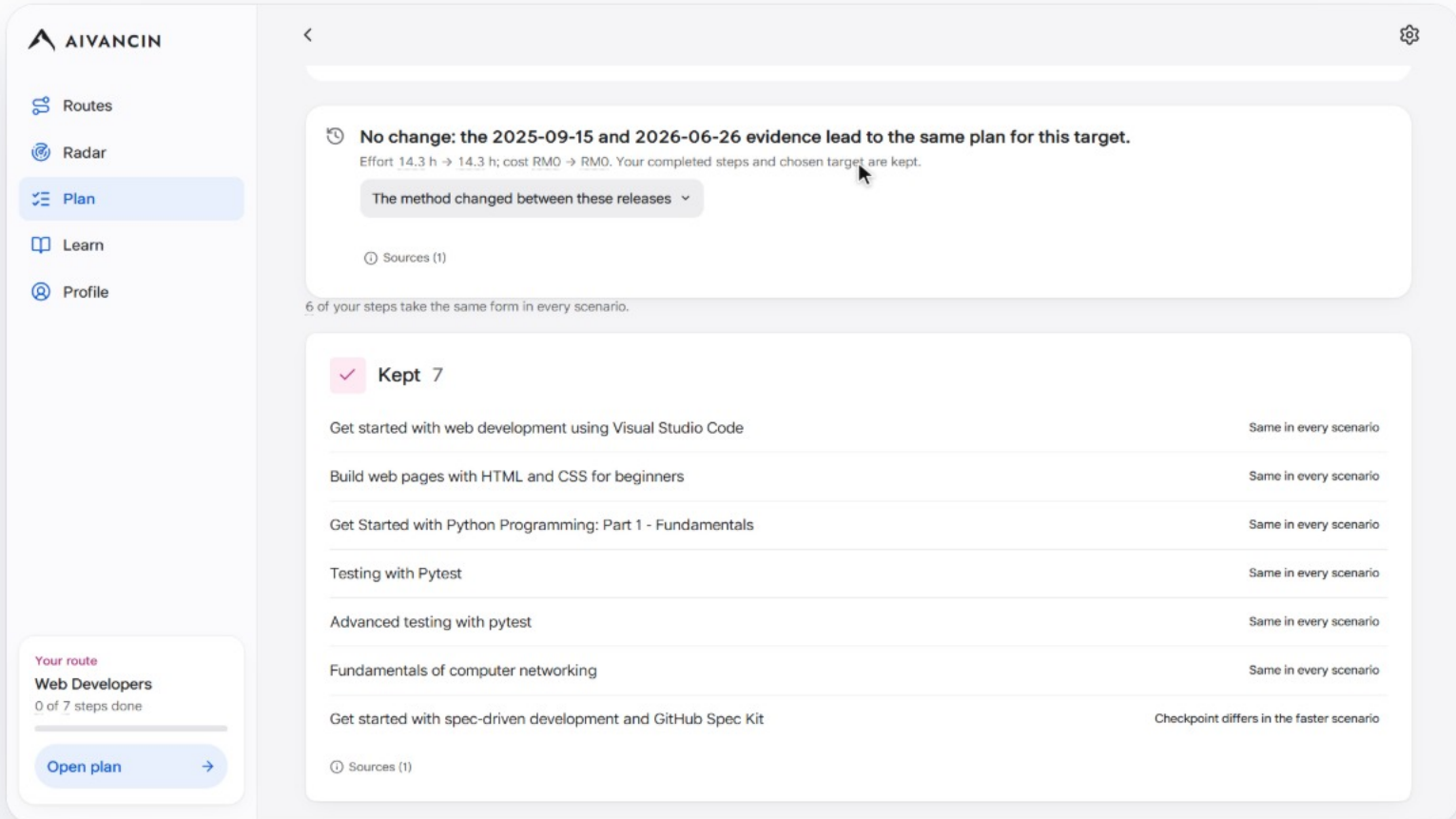
Four weeks of free courses, sized to your hours and budget, prerequisites first. Start here names the exact first unit, setup and what you will make.



Live prototype · What changed

New evidence, your decision

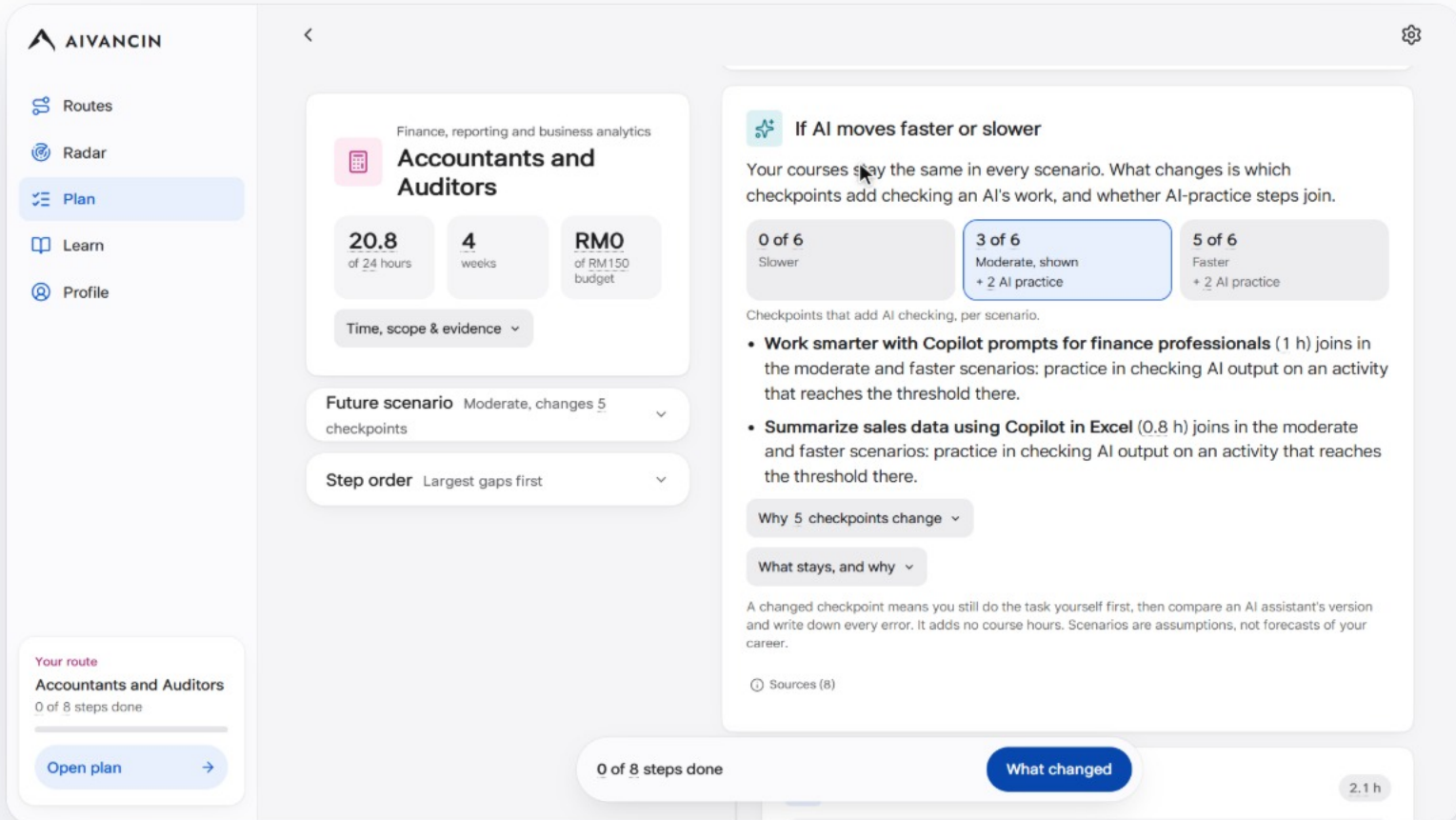
New AI-usage data arrives: the app compares your plan under both releases, with your work and progress fixed. Here nothing changes, and it says so. You decide.



Prototype · Alternative assumptions

What changes in the plan

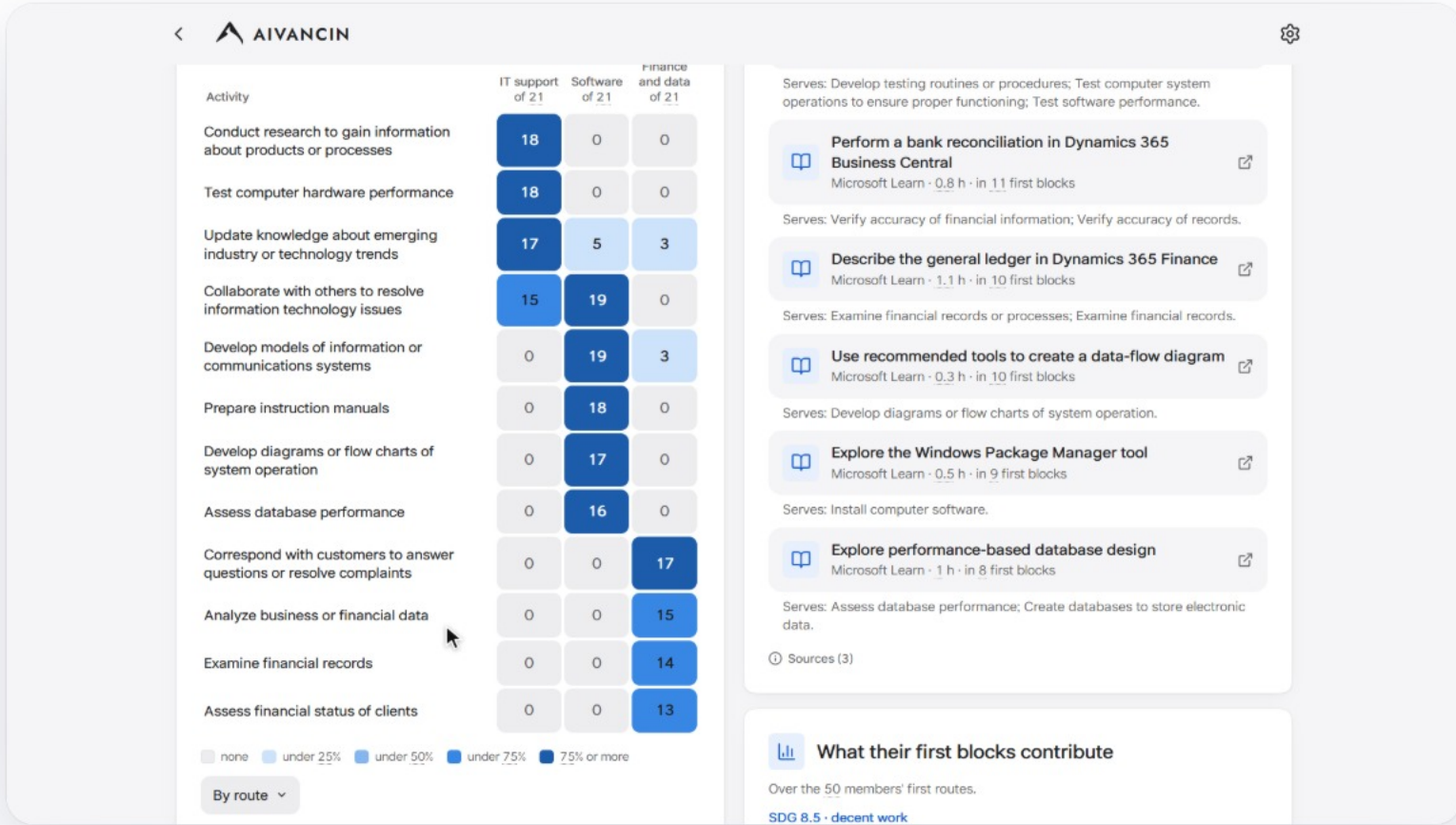
Priya's accounting route: zero, three and five of six checkpoints add AI-checking practice under slower, moderate and faster adoption. Her foundations stay first in all three.



Live prototype · Career centres

A cohort view

Career centres see shared skill gaps across 60 synthetic graduates, by route family. Counts only, never a score; nobody's routes change.



What Hafiz walks away with

Routes built on work he already does, and a plan to start.

Routes

3 graduate roles

Network support, web development and programming, each built on work he confirmed.

Still needed

What each requires

Missing skills, entry barriers such as a degree, and how AI is changing the work.

Plan

A free 4-week plan

Eight hours a week, prerequisites first, starting with one exact first task.

He chooses the target. Every number shows its source.

Domain One must-haves

All six run in the prototype.

1

Skills profile intake

Four questions in your own words. AI suggests work activities; you confirm each one.

2

AI-trend signal layer

Four dated AI-usage releases, US job outlook (BLS) and Malaysian data (DOSM), each with its source.

3

Future-role matching

Roles ranked by shared work, with entry requirements and skill gaps.

4

Reskilling pathway

An ordered block of verified free courses, sized to your hours and budget.

5

SDG mapping

Every role and plan step tagged to SDG targets with a reason, plus your path's contribution.

6

Handling uncertainty

Slower, moderate and faster AI scenarios; every projection labelled an estimate.

How it works

A small model, checked at every step.

1. **You describe your work**
   plain sentences, your own words
2. **AI suggests work activities**
   fine-tuned MiniLM on our server; 2,087 O*NET activities
3. **You confirm what is true**
   nothing is accepted silently
4. **Matching and plans by fixed rules**
   same input, same result; no generated text
5. **Results with sources and dates**
   O*NET, Anthropic Economic Index, BLS, DOSM

RunPod trained

Fine-tuned once on an **RTX 4090** with O*NET task-to-activity pairs. Shipped because it passed a gate fixed before training: **45 of 60** held-out descriptions, against 39. Cost US$0.42.

Replit live

The prototype runs at **aivancin--jadfahmi.replit.app** from bundled, dated, hashed public data. No API keys, no per-request cost.

Evidence

What we measured on synthetic profiles.

32 : 20

relevant top-five roles out of 50: confirmed AI suggestions versus keyword search

Identical synthetic descriptions

Pass

45/60

held-out descriptions matched correctly after fine-tuning

39 before; test fixed before training

Shipped

7/12

test profiles with two complete routes and a free plan

58%, against our 80% target; keywords 2/12, job titles 0/12

Not yet met

Hardest risk: missing someone's work, even in supported fields. **Fallback:** users confirm every suggestion and can search for missed work; incomplete routes say so. Never an invented job or course.

Predeclared synthetic set 8; first run and final rerun 7/12. No real-user outcome validation. 354 Python tests and 14 Node tests pass (23 Oct). Details in the README.

SDG impact

From the work you do to decent work.

Targets	SDG 8.5 SDG 4.4 Decent work for young people; relevant skills for employment. Supporting: 4.3 free learning, 8.2 working with AI.
Baseline	**40.7%** of employed graduates aged 25 to 34 in semi- or low-skilled jobs (751,100; DOSM, Q2 2026)
KPI	Route-and-block coverage: the share of test profiles that get a usable plan (two relevant routes, sourced requirements, a free four-week block). **Today: 7 of 12 (58%)**
Phase 2 target	**At least 80%** on a fresh profile set fixed in advance; then trials with real underemployed graduates

AIVANCIN

You choose.
AI suggests.

Live prototype aivancin--jadfahmi.replit.app

Code github.com/AIVANCIN/aivancin

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